

Study of Linear Accelerators: Imaging and Beam Optimization

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Introduction

The aim of this thesis is to study Linear Accelerators (Linacs) on the context of their applications in external beam radiotherapy (EBRT). Especially, EBRT based on hadrons such as protons and carbon-ions is focused on. A major challenge is real time verification of beam dose deposition. Conventional imaging techniques lack temporal resolution. To address this, we use prompt gammas, which are emitted instantaneously during nuclear interactions between the incident proton beam and target nuclei, providing real-time information on beam penetration and interaction depth. At the same time, equipped with a dependable imaging system, we investigate the quality of the beam using several beam parameters. We study beam dynamics and beam transport in Linacs to do this. By combining imaging and beam optimization, we would have robust, real-time clinically viable proton therapy system.

Prompt Gamma Imaging

The idea behind using proton beams is the presence of the bragg peak, which ensure accurate energy deposition in the target material.

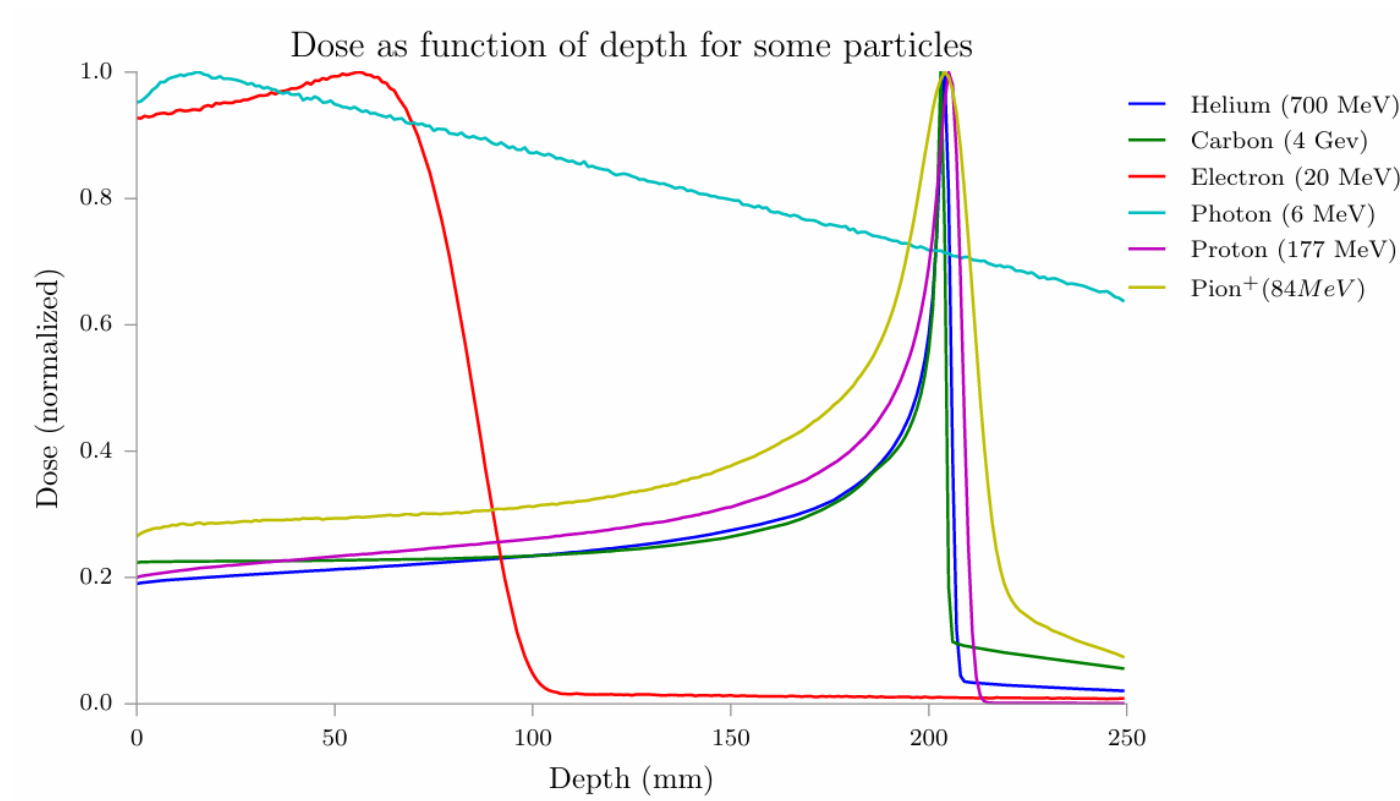


Figure 1. Energy deposition range for various particles

The range of a particle is defined as,

$$r = \int_0^{E_{inc}} \frac{1}{\rho S(E)} dE \quad (1)$$

where, $S(E)$ is the stopping power and is given by the bethe-block formula for proton.

$$-\frac{dE}{dx} = \frac{4\pi n_e e^4}{m_e c^2 \beta^2} \ln \left(\frac{2m_e c^2 \beta^2}{I} \right) \quad (2)$$

The reason why prompt gammas (PGs) are used for imaging is because the PG fall-off is within 5cm of the distal fall-off.

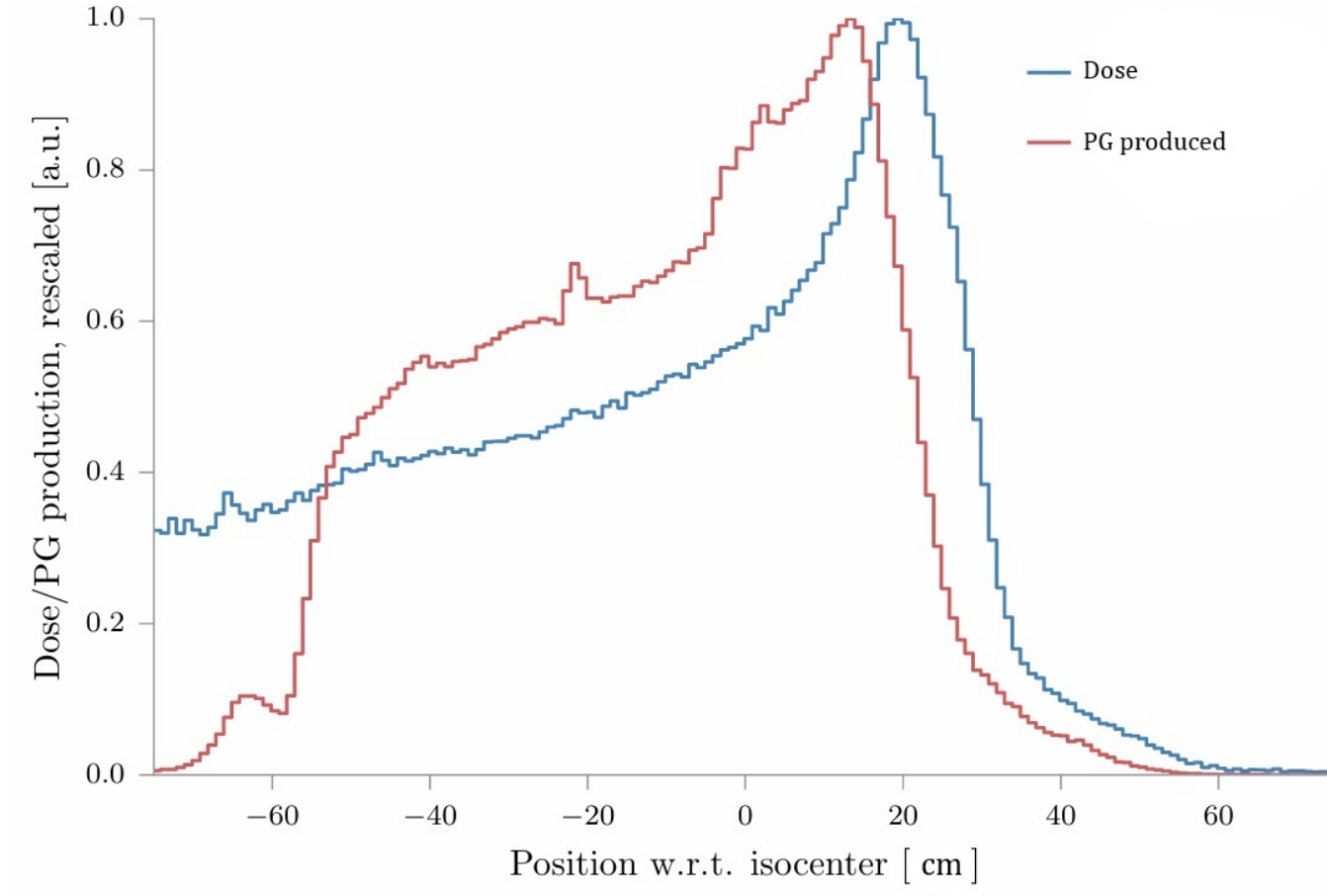


Figure 2. Dose and PG distribution

Now, once we have the PG emission, we use a Compton Camera along with collimation devices to image the beam. The Compton Camera has 2 layers called the Scatterer and the Absorber layer.

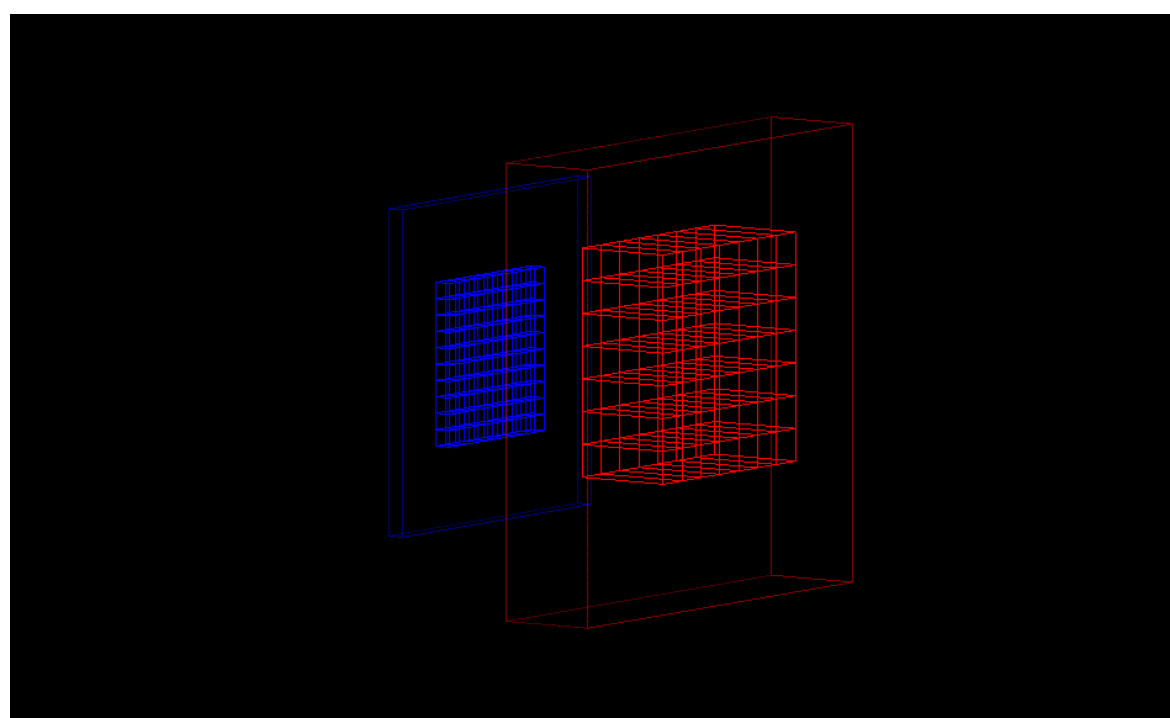


Figure 3. A Geant4 geometry of the compton camera

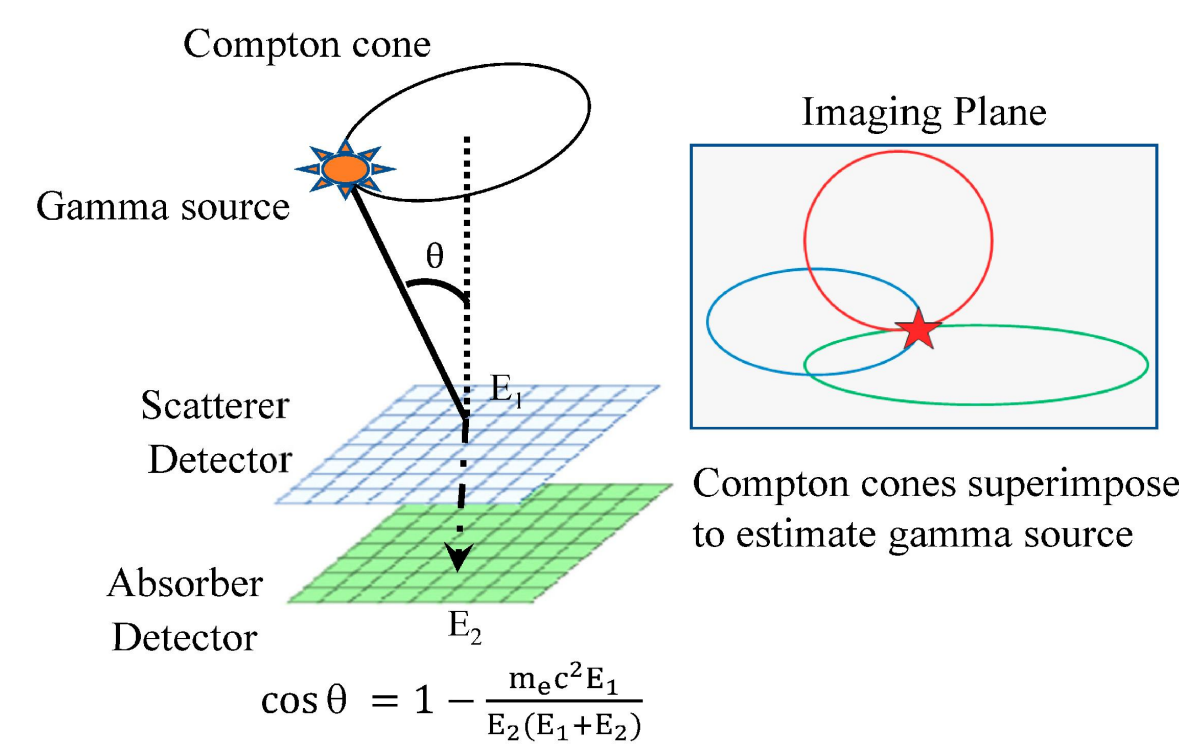


Figure 4. Working of a compton camera

It has been observed that the image is filled with noise arising from secondary neutron production. Without any spatial filtering, detectors would integrate from all direction leading to a bad image. Collimators are introduced to restrict the angular acceptance of incoming gamma rays, effectively acting as a spatial filter. We evaluate 2 collimators along with our gamma detectors: Multi-Slit Collimators and Knife-Edge Collimators.

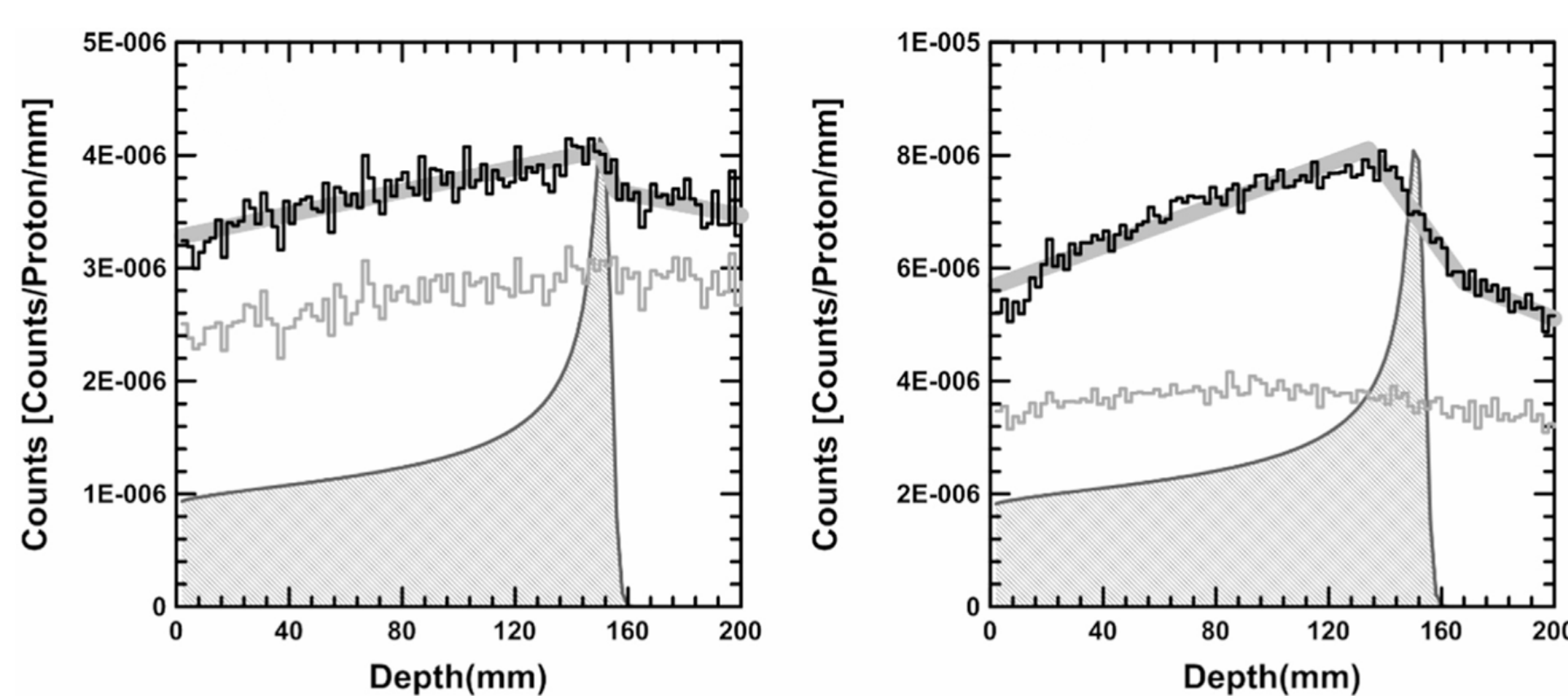


Figure 5. Particle counts from different collimator based imaging systems, Left is Multi-Slit and Right is Knife-Edge

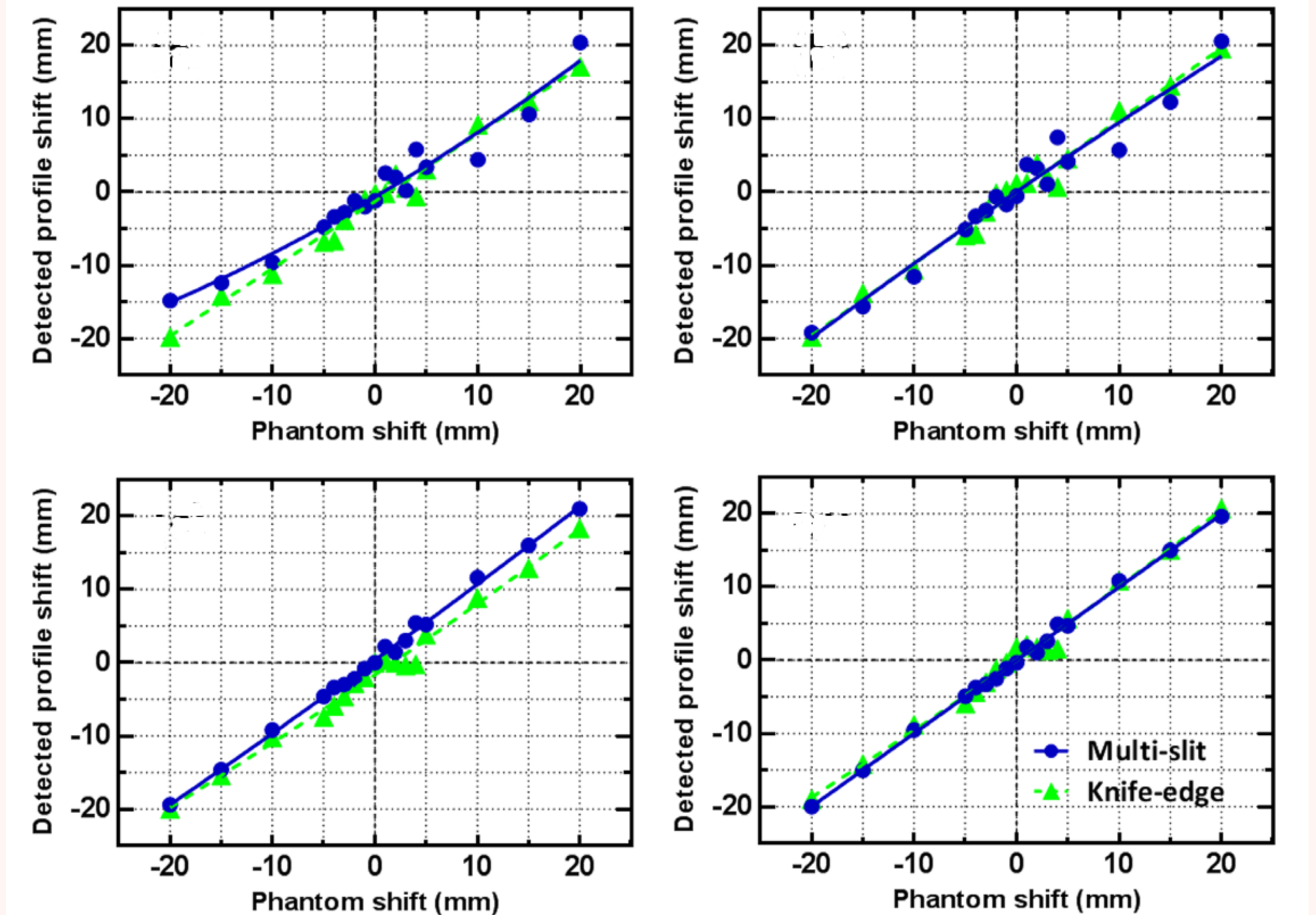


Figure 6. Range shifts vs Phantom shifts for different collimators

Beam Dynamics Studies

Beam parameters and their equation are studied and it is seen they are equally important to ensure beam quality. Accurate dose delivery requires a well transported, shaped and controlled beam. Beam Emittance is one of the most important beam parameters.

$$\epsilon_x = \sqrt{\langle x^2 \rangle \langle x'^2 \rangle - \langle x x' \rangle^2} \quad (3)$$

where $\langle x^2 \rangle$ and $\langle x'^2 \rangle$ are the position and angular variance. While, $\langle x x' \rangle^2$ is the correlation between them.

Emittance is defined as the area occupied by the phase space of the beam. Beamline elements such as quadrupole magnets need to be tuned for optimal quadrupole strengths. We study two optimization algorithms for this; Gradient based Optimization and Genetic Algorithm.

Beam Optimization

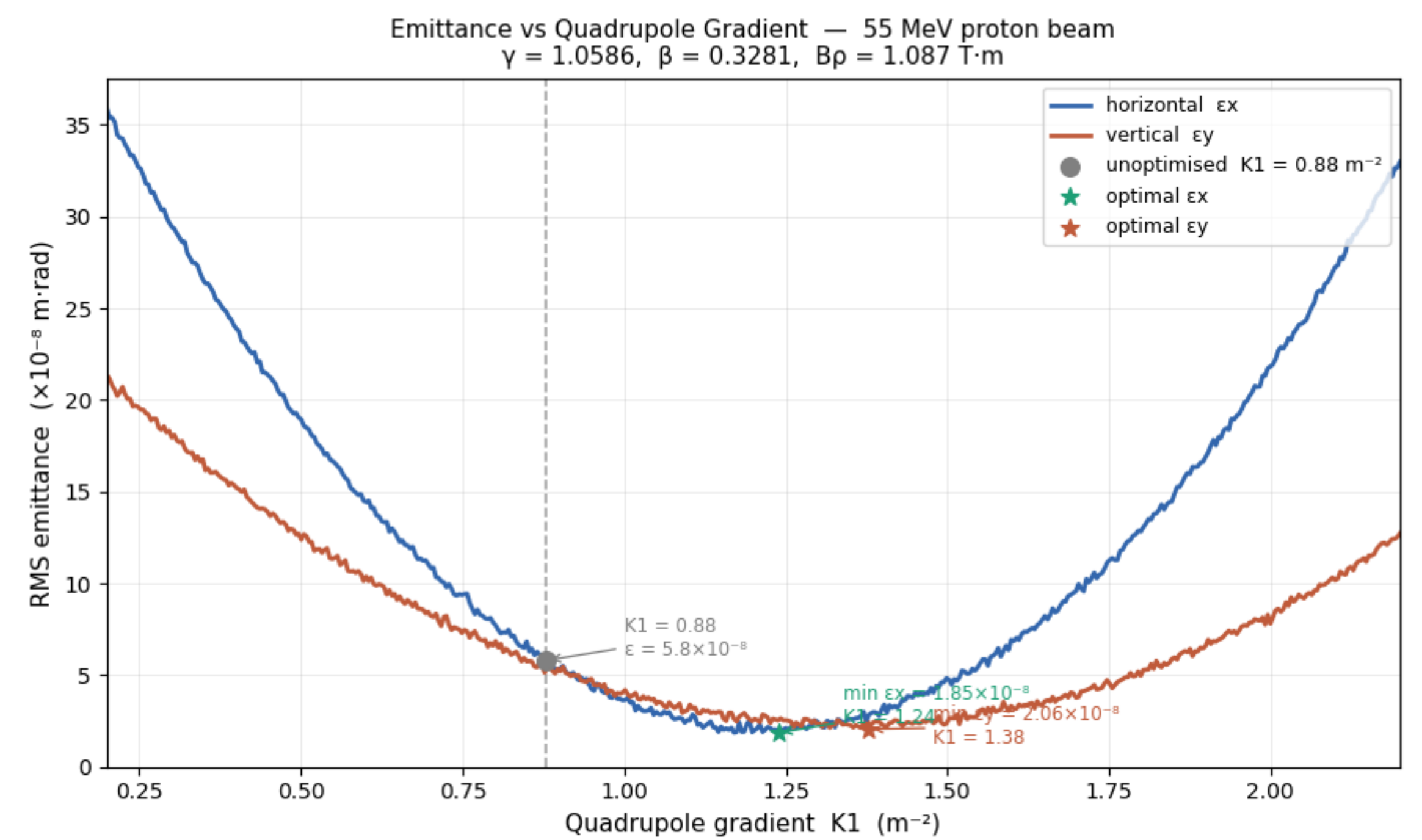


Figure 7. Emittance vs quadrupole strength trend as seen using the gradient based optimization

Beam Model

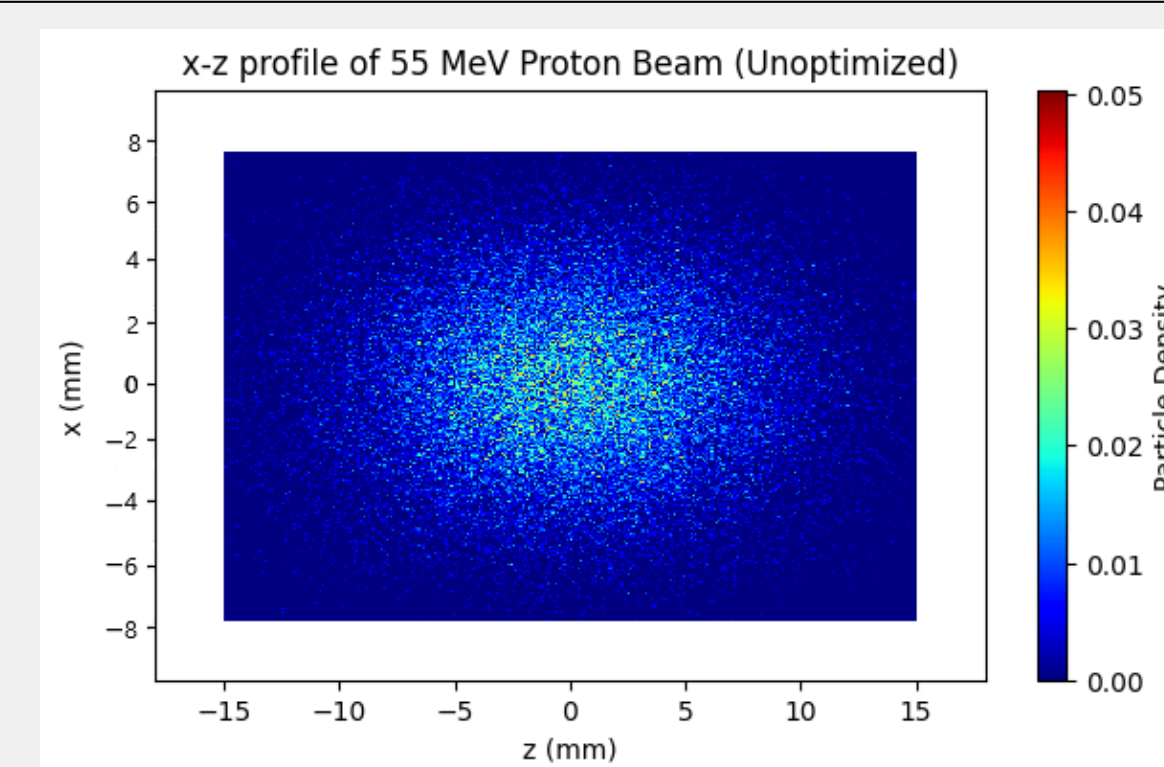


Figure 8. Beam Profile for an unoptimized 55 MeV beam

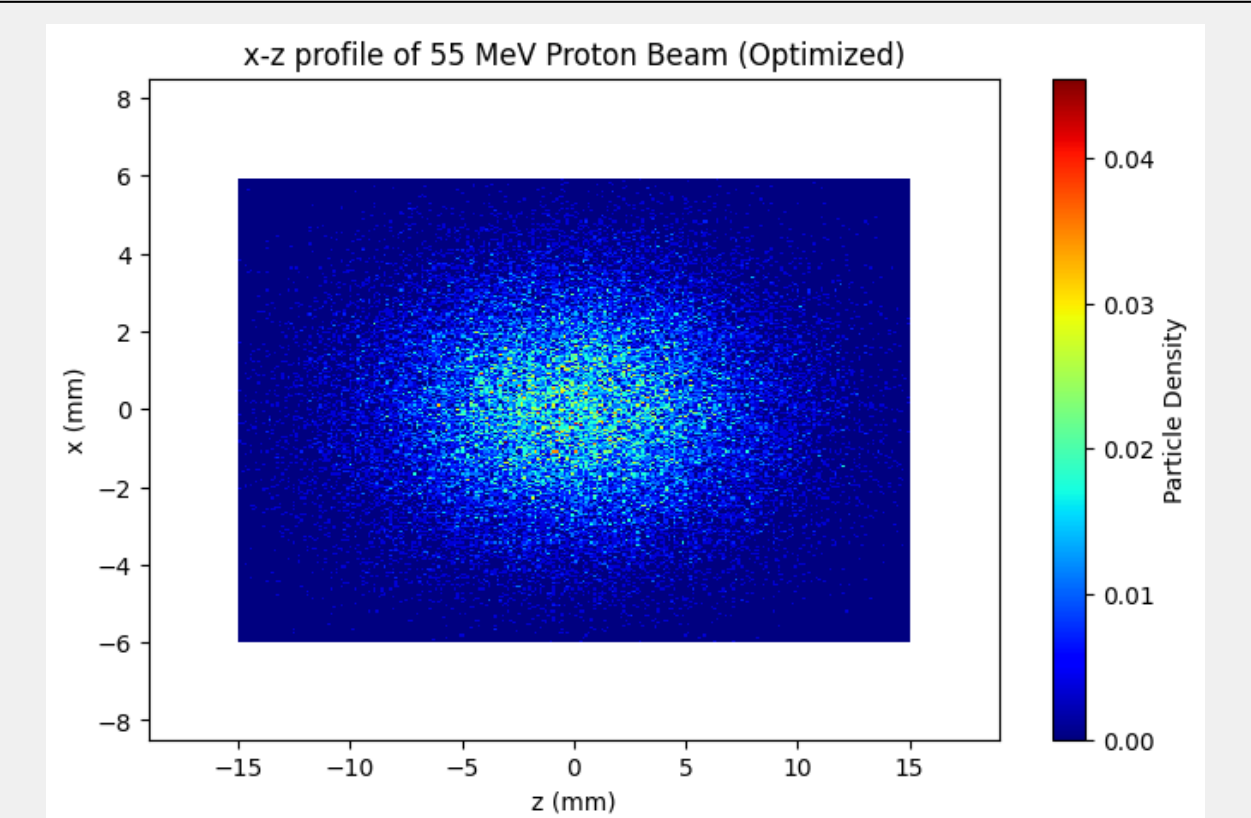


Figure 9. Beam Profile after optimization

We see that the Beam size has reduced after modelling it after optimized parameters. Thus, achieving what we set out to do!

References

- [1] Brent Huisman. Accelerated prompt gamma simulations for proton therapy. 2019.
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- [3] Shogo Sato, Hiromu Yokokawa, Mana Hosobuchi, and Kataoka Jun. A simulation study of in-beam visualization system for proton therapy by monitoring scattered protons. *Front. Med. (Lausanne)*, 10:1038348, jul 2023.